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1.7.3 Arithmetical Operations with Significant Figures

(i) Addition and subtraction

In addition and subtraction, the final result should retain as many decimal places as there are in the number with the smallest number of decimal places.

Example:

1. 3.1 + 1.780 + 2.046 = 6.926

   Here the least number of significant digits after the decimal is one. Hence the result will be 6.9.

2. 12.637 – 2.42 = 10.217

   Here the least number of significant digits after the decimal is two. Hence the result will be 10.22

(ii) Multiplication and Division

In multiplication or division, the final result should retain as many significant figures as there are in the original number with smallest number of significant figures.

Example:

1. 1.21 × 36.72 = 44.4312 = 44.4

   Here the least number of significant digits in the measured values is three. Hence the result when rounded off to three significant digits is 44.4

2. 36.72 ÷ 1.2 = 30.6 = 31

   Here the least number of significant digits in the measured values is two. Hence the result when rounded off to significant digit becomes 31.

1.8 DIMENSIONAL ANALYSIS

1.8.1 Dimension of Physical Quantities

All the derived physical quantities can be expressed in terms of some combination of the seven fundamental or base quantities. These base quantities are known as dimensions of the physical world, and are denoted with square bracket [ ]. The three dimensions in mechanics are [L] for length, [M] for mass and [T] for time. In electricity, [A] is the dimension of electric current. In thermodynamics, [K] is the dimension for the temperature. In optics [cd] or [Φ] is the dimension for luminous intensity. The dimension of amount of substance is [mol].

The dimensions of a physical quantity are the powers to which the units of base quantities are raised to represent a derived unit of that quantity.

For example,

$$velocity = \frac{Displacement}{time} = \frac{[L]}{[T]} = [M^0LT^{-1}].$$

Hence the dimensions of velocity are 0 in mass, 1 in length and -1 in time.

Dimensional formula and equation

Dimensional formula is an expression which shows how and which of the fundamental units are required to represent the unit of a physical quantity.

For example, $[M^0LT^{-2}]$ is the dimensional formula of acceleration.

When the dimensional formula of a physical quantity is expressed in the form of an equation, such an equation is known as the dimensional equation.